

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-30944890})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}5851 + \mathbf{Z}(2467 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 30944890 = 0$, of discriminant -123779560 . Class group invariants $(10 \ 10 \ 10 \ 2)$, class number 2000; the three stored generator ideals (HNF) are $\begin{pmatrix} 199 & 159 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 193 & 99 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 157 & 81 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 6317)y^3 + (-6s - 354679)y^2 + (863s + 3255502)y + (-6552s + 9323580)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 12635y^8 - 721992y^7 + 78069741y^6 - 4097539650y^5 \\ & + 114611694049y^4 - 2106485720276y^3 + 29464343022134y^2 - 2892 \\ & 42220002960y + 1415353167138960 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + \left(-\frac{418447}{6857258011452254251}\right)s, \quad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5760833231840 = 2^5 \cdot 5 \cdot 23 \cdot 1565443813$:

$$\begin{pmatrix} 720104153980 & 635798831956 & 88527053297 & 519937678167 & 482354874216 & 505693212416 & 421856546838 & 631834141300 & 375408409468 & 620188178820 \\ 0 & 4 & 0 & 0 & 2 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 314 factors with signed exponents of absolute value up to 277861696774991; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 1 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 6317)y^3 + (-6s - 354679)y^2 + (863s + 3255502)y + (-6552s + 9323580)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 12635y^8 - 721992y^7 + 78069741y^6 - 4097539650y^5 + 114611694049y^4 - 2106485720276y^3 + 29464343022134y^2 - 289242220002960y + 1415353167138960$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849} \right) s, \quad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $43807961796496 = 2^4 \cdot 23^2 \cdot 14891 \cdot 347579$:

$$\begin{pmatrix} 476173497788 & 202281566250 & 263495035516 & 385649339052 & 30691944248 & 367716659449 & 240380892912 & 18581880099 & 15112052413 & 11504705617 \\ 0 & 2 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 46 & 13 & 24 & 15 & 4 & 31 & 27 & 29 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 267 factors with signed exponents of absolute value up to 234989190205813; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 6317)y^3 + (-6s - 354679)y^2 + (863s + 3255502)y + (-6552s + 9323580)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 12635y^8 - 721992y^7 + 78069741y^6 - 4097539650y^5 + 114611694049y^4 - 2106485720276y^3 + 29464343022134y^2 - 289242220002960y + 1415353167138960$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + \left(-\frac{1403}{10368641602001000}\right) s, \quad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $55535605473200 = 2^4 \cdot 5^2 \cdot 83 \cdot 1672759201$:

$$\begin{pmatrix} 1388390136830 & 231626424108 & 290168754980 & 267053079090 & 26361350772 & 314677876133 & 1158620142392 & 575751006125 & 724748895157 & 237069318779 \\ 0 & 2 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 10 & 4 & 3 & 8 & 0 & 6 & 2 & 2 \\ 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 262 factors with signed exponents of absolute value up to 273806086478056; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 350y^3 - 36sy^2 - 2232216y + 1296s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 700y^8 - 4341932y^6 + 38542026240y^4 + 2095258694976y^2 + 51975532362240$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 2 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + \left(-\frac{418447}{6857258011452254251}\right) s, \quad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10164350346390 = 2 \cdot 3^2 \cdot 5 \cdot 23 \cdot 30089 \cdot 163193$:

$$\begin{pmatrix} 3388116782130 & 738974532968 & 852479789800 & 602058177221 & 2028816217100 & 1254925282458 & 3218844804028 & 1908532760831 & 1990844299489 & 556095569316 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 202 factors with signed exponents of absolute value up to 194534772491075; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 350y^3 - 36sy^2 - 2232216y + 1296s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 700y^8 - 4341932y^6 + 38542026240y^4 + 2095258694976y^2 + 51975532362240$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 2 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849} \right) s, \quad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $373678321798385 = 5 \cdot 43 \cdot 47^2 \cdot 2089 \cdot 376639$:

$$\begin{pmatrix} 7950602591455 & 7676350054770 & 2386961699267 & 3729135836463 & 1948513309673 & 1209063530838 & 580293353483 & 3686019349724 & 6655536286429 & 5484826714423 \\ 0 & 47 & 42 & 42 & 1 & 16 & 0 & 2 & 16 & 10 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 220 factors with signed exponents of absolute value up to 73660983590830; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 350y^3 - 36sy^2 - 2232216y + 1296s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 700y^8 - 4341932y^6 + 38542026240y^4 + 2095258694976y^2 + 51975532362240$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 2 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + \left(-\frac{1403}{10368641602001000}\right)s, \quad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $69376616559000 = 2^3 \cdot 3^4 \cdot 5^3 \cdot 23 \cdot 41^2 \cdot 22153$:

$$\begin{pmatrix} 3133541850 & 2301356076 & 227538962 & 915124849 & 800847911 & 1340373220 & 2072635704 & 1693879079 & 2236123846 & 74397667 \\ 0 & 123 & 0 & 87 & 10 & 45 & 66 & 66 & 57 & 113 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 1 & 4 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 207 factors with signed exponents of absolute value up to 391557511845473; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (-s + 1184)y^3 + (29s - 134990)y^2 + (-575s + 735135)y + (8035s + 15013550)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} + 2368y^8 - 269980y^7 + 33817016y^6 - 2084432840y^5 + 81574 \\ & 375770y^4 - 1692216124700y^3 + 21139466582175y^2 - 26386454766400 \\ & 0y + 2223246689492750 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + \left(-\frac{418447}{6857258011452254251}\right)s, \quad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $19872590 = 2 \cdot 5 \cdot 1381 \cdot 1439$:

$$\begin{pmatrix} 19872590 & 7218944 & 6505022 & 5755294 & 4812106 & 14270977 & 18060912 & 3746316 & 16134451 & 8561617 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 270 factors with signed exponents of absolute value up to 9465277282274512; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (-s + 1184)y^3 + (29s - 134990)y^2 + (-575s + 735135)y + (8035s + 15013550)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} + 2368y^8 - 269980y^7 + 33817016y^6 - 2084432840y^5 + 81574 \\ & 375770y^4 - 1692216124700y^3 + 21139466582175y^2 - 26386454766400 \\ & 0y + 2223246689492750 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849} \right) s, \quad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1078714338834340 = 2^2 \cdot 5 \cdot 13^2 \cdot 23 \cdot 179 \cdot 77519129$:

$$\begin{pmatrix} 41489013032090 & 17509552991324 & 20516276877834 & 20374718940929 & 17464744551231 & 17054968657267 & 19338136669172 & 37522605803983 & 18402342521393 & 16514873231059 \\ 0 & 2 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 13 & 0 & 2 & 11 & 1 & 11 & 4 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 275 factors with signed exponents of absolute value up to 11667307327937480; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (2 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (-s + 1184)y^3 + (29s - 134990)y^2 + (-575s + 735135)y + (8035s + 15013550)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} + 2368y^8 - 269980y^7 + 33817016y^6 - 2084432840y^5 + 81574 \\ & 375770y^4 - 1692216124700y^3 + 21139466582175y^2 - 26386454766400 \\ & 0y + 2223246689492750 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + \left(-\frac{1403}{10368641602001000}\right)s, \quad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $303301286209190 = 2 \cdot 5 \cdot 30330128620919$:

$$\begin{pmatrix} 303301286209190 & 144535334070024 & 293330223000494 & 257268800455997 & 44281878398431 & 259087006898581 & 203611577663471 & 161894452910022 & 257528713041066 & 294738176462577 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 264 factors with signed exponents of absolute value up to 8353534204799890; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2731y^3 - 16sy^2 - 1232700y + 2720s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 5462y^8 + 4992961y^6 + 1188884440y^4 - 1173893935600y^2 + 228942674176000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ 1 \ -2 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + \left(-\frac{418447}{6857258011452254251}\right)s, \quad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')^J = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $204896335075288 = 2^3 \cdot 17^2 \cdot 257 \cdot 1229 \cdot 280583$:

$$\begin{pmatrix} 3013181398166 & 2743551611550 & 1602475580882 & 2147121121112 & 2583422040399 & 2888272425438 & 2491093196943 & 30956552698 & 2623692094934 & 1381786880632 \\ 0 & 2 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 17 & 1 & 0 & 4 & 10 & 4 & 16 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 280 factors with signed exponents of absolute value up to 14775808026; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2731y^3 - 16sy^2 - 1232700y + 2720s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 5462y^8 + 4992961y^6 + 1188884440y^4 - 1173893935600y^2 + 228942674176000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ 1 \ -2 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849} \right) s, \quad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a')^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $121258264420080 = 2^4 \cdot 3^2 \cdot 5 \cdot 168414256139$:

$$\begin{pmatrix} 10104855368340 & 9345938218092 & 1736754698582 & 155971325962 & 8746527377753 & 8245274824972 & 7634551891357 & 4228615021960 & 1329977864360 & 7969642513744 \\ 0 & 6 & 2 & 1 & 3 & 2 & 4 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 295 factors with signed exponents of absolute value up to 9082335405; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2731y^3 - 16sy^2 - 1232700y + 2720s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 5462y^8 + 4992961y^6 + 1188884440y^4 - 1173893935600y^2 + 228942674176000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ 1 \ -2 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + \left(-\frac{1403}{10368641602001000}\right) s, \quad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $76340748047450 = 2 \cdot 5^2 \cdot 157 \cdot 9724936057$:

$$\begin{pmatrix} 15268149609490 & 4015539492132 & 15143472830740 & 4176181107517 & 8580765040692 & 4650534239978 & 10347153147842 & 12899515824097 & 5540278246869 & 7670252068737 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 4 & 0 & 3 & 4 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 292 factors with signed exponents of absolute value up to 9371789712; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 796)y^3 + (29s - 86202)y^2 + (-259s + 3391035)y + (-2145s - 26272620)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 1596y^8 - 175588y^7 + 38705384y^6 - 1998146584y^5 \\ & + 54988508514y^4 - 958554568660y^3 + 14254583550895y^2 - 1437995 \\ & 71195500y + 832628774176650 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + \left(-\frac{418447}{6857258011452254251}\right) s, \quad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $51391995874488 = 2^3 \cdot 3^2 \cdot 569 \cdot 1254442391$:

$$\begin{pmatrix} 4282666322874 & 2151242738856 & 2953408694610 & 2985711139966 & 3605334430085 & 3463802993073 & 2217459160528 & 1187121733047 & 1684702554658 & 1574334602152 \\ 0 & 6 & 2 & 1 & 4 & 4 & 1 & 2 & 4 & 2 \\ 0 & 0 & 2 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 287 factors with signed exponents of absolute value up to 3901317161540652; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 3 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 796)y^3 + (29s - 86202)y^2 + (-259s + 3391035)y + (-2145s - 26272620)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 1596y^8 - 175588y^7 + 38705384y^6 - 1998146584y^5 \\ & + 54988508514y^4 - 958554568660y^3 + 14254583550895y^2 - 1437995 \\ & 71195500y + 832628774176650 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849} \right) s, \quad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $43176275936000 = 2^8 \cdot 5^3 \cdot 149 \cdot 2029 \cdot 4463$:

$$\begin{pmatrix} 53970344920 & 45391638180 & 42672553330 & 11800538470 & 20085537740 & 11917797526 & 25664641930 & 22929394091 & 7784409348 & 30292019252 \\ 0 & 4 & 0 & 2 & 0 & 1 & 1 & 0 & 1 & 3 \\ 0 & 0 & 10 & 0 & 0 & 9 & 7 & 2 & 9 & 1 \\ 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 10 & 3 & 2 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 286 factors with signed exponents of absolute value up to 2861328727268913; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 796)y^3 + (29s - 86202)y^2 + (-259s + 3391035)y + (-2145s - 26272620)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 1596y^8 - 175588y^7 + 38705384y^6 - 1998146584y^5 \\ & + 54988508514y^4 - 958554568660y^3 + 14254583550895y^2 - 1437995 \\ & 71195500y + 832628774176650 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + \left(-\frac{1403}{10368641602001000}\right)s, \quad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $35172728358046 = 2 \cdot 1979 \cdot 8886490237$:

$$\begin{pmatrix} 35172728358046 & 2763587212181 & 24506509926714 & 31604259146119 & 32995168233535 & 14910789804599 & 10236947070795 & 10276446727203 & 15105446998558 & 19110838546860 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 290 factors with signed exponents of absolute value up to 1202361373399609; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (3 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1361y^3 - 2sy^2 + 205690y + 200s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2722y^8 + 2263701y^6 - 436108620y^4 + 17552464100y^2 + 1237795600000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ -1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + \left(-\frac{418447}{6857258011452254251}\right)s, \quad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $117296673946304 = 2^6 \cdot 1832760530411$:

$$\begin{pmatrix} 3665521060822 & 1031049478788 & 2881010444528 & 2855985938460 & 1287594759918 & 1936714745841 & 1291932329037 & 1705565258488 & 838021523557 & 3195614264611 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 300 factors with signed exponents of absolute value up to 515249827265; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (0 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1361y^3 - 2sy^2 + 205690y + 200s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2722y^8 + 2263701y^6 - 436108620y^4 + 17552464100y^2 + 1237795600000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ -1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849} \right) s, \quad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $292322561024000 = 2^{13} \cdot 5^3 \cdot 13^2 \cdot 883 \cdot 1913$:

$$\begin{pmatrix} 1756746160 & 1233405680 & 1697379940 & 207087290 & 257914106 & 680054070 & 1279015750 & 190737176 & 803020491 & 195358384 \\ 0 & 520 & 160 & 250 & 116 & 8 & 184 & 481 & 63 & 429 \\ 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 10 & 6 & 8 & 2 & 7 & 6 & 7 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 295 factors with signed exponents of absolute value up to 157065219523; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1361y^3 - 2sy^2 + 205690y + 200s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2722y^8 + 2263701y^6 - 436108620y^4 + 17552464100y^2 + 1237795600000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ -1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + \left(-\frac{1403}{10368641602001000}\right)s, \quad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $53104303260000 = 2^5 \cdot 3^2 \cdot 5^4 \cdot 1049 \cdot 281243$:

$$\begin{pmatrix} 17701434420 & 2475966390 & 2897088930 & 12605418090 & 9610582418 & 3155480775 & 17238565915 & 9913437973 & 17652246375 & 11093366319 \\ 0 & 5 & 0 & 0 & 3 & 4 & 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 30 & 28 & 24 & 18 & 14 & 0 & 13 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 309 factors with signed exponents of absolute value up to 403594772202; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.